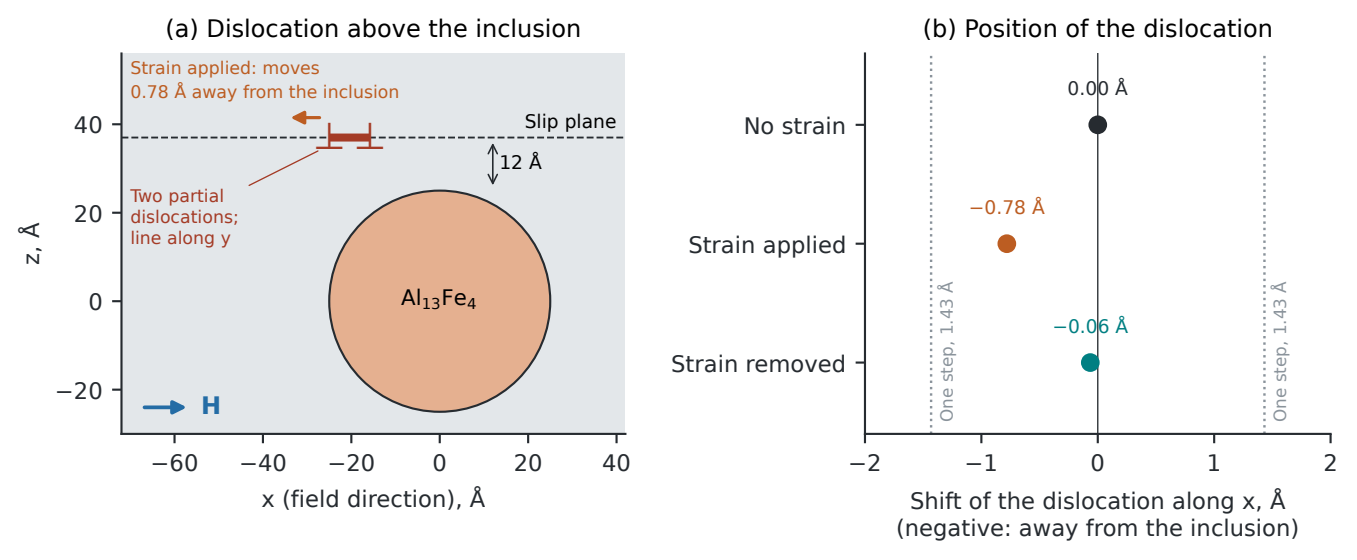
(a) Dislocation above the inclusion
Strain applied: moves
0.78 Å away from the inclusion
Slip plane
12 Å
Two partial
dislocations;
line along y
$Al_{13}Fe_4$
H
z, Å
x (field direction), Å
40
20
0
−20
−60
−40
−20
0
20
40
(b) Position of the dislocation
No strain
Strain applied
Strain removed
0.00 Å
−0.78 Å
−0.06 Å
One step, 1.43 Å
One step, 1.43 Å
−2
−1
0
1
2
Shift of the dislocation along x, Å
(negative: away from the inclusion)